

Introduction to Jitter and Phase Noise

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Presentation Topics

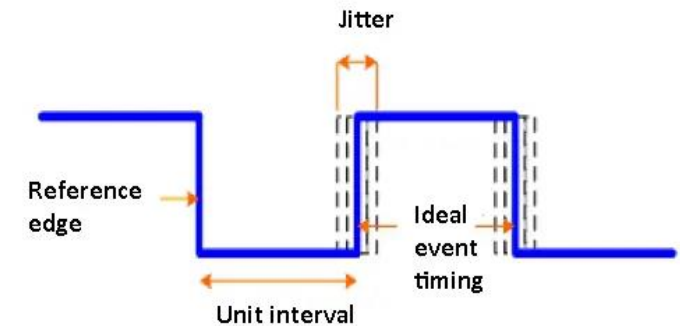
1. Introduction to the main topics discussed
2. Motivation for the chapter
3. Noise & Its types
4. Device Noise and Propagation
5. Power and Spectral Density
6. Jitter & Phase Noise : definition and types
7. Jitter vs Phase noise
8. Types of Jitter
9. Trade-offs in Design
10. Basic Phase noise mechanisms.
11. Effect of Jitter & PN on Performance

Motivation – Why is this chapter Important ?

- In the Last chapter, we understood that phase noise and power are inversely related. Balancing between the two is very important.
- The design of PLL deal with jitter and phase noise.
- That's why this chapter focuses on jitter and phase noise before diving into the detailed design of ring and LC oscillators.

Intro to Jitter and Phase Noise

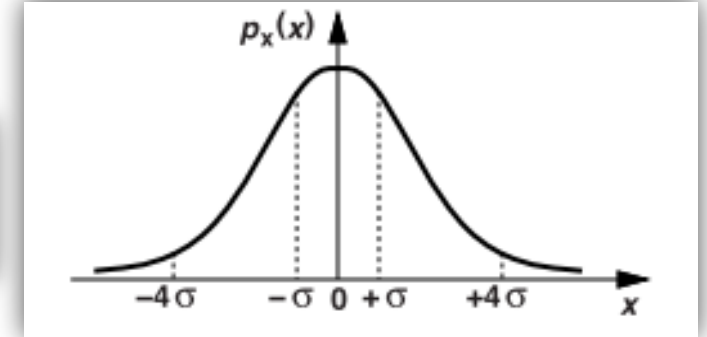
- **Jitter** happens when the expected zero-crossing of the signal is shifted slightly from where it should be.
 - It is a Time-Domain Phenomenon..!



- **Phase Noise** is the random fluctuations in the phase (or zero crossings) of an oscillator o/p
 - It is a Frequency-Domain Phenomenon..!

We will dive into detailed comparison later in the presentation

Noise: Concepts & Types



- **Noise is random:**

- Its value at any instant (in time) can't be predicted.

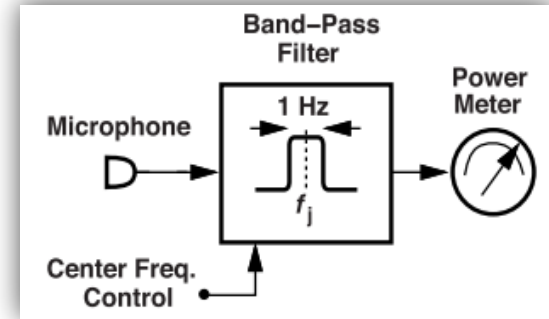
Examples: traffic sounds, people talking in a restaurant, river flow.

- **Time Domain View :**

- The **probability density function (PDF)** tells us how often different noise values (or noise amplitudes) can happen.
- if the noise is generated by many sources independently, we expect its PDF to be a Gaussian distribution. (bell-shaped)
- curve is centered at the mean (μ), and its spread is determined by the **standard deviation (σ)**
 - Peak value (or amplitude) rarely exceeds $\pm 4\sigma$

- **Frequency Domain View:**

- Generally, noise is studied in Frequency-domain.
- Since Noise is Random and is spread across the freq-range ,
Fourier Transform of the signal will not result in a clean Frequency chart.

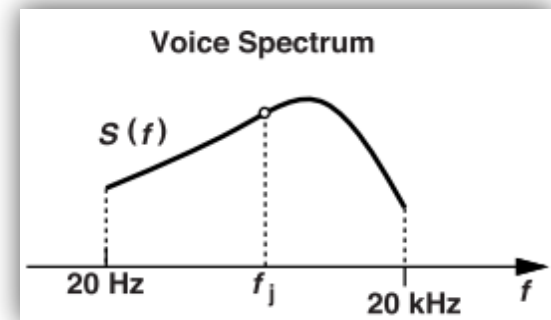


How is this solved ??

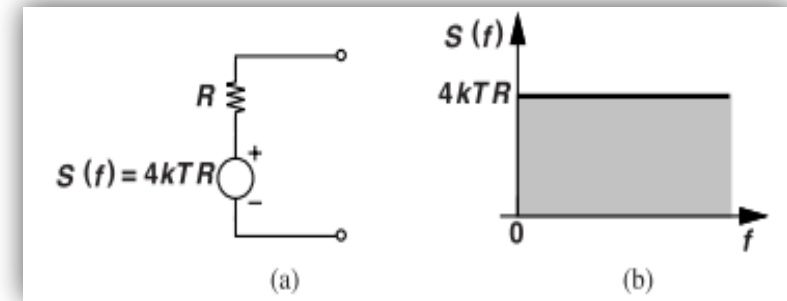
- We analyze noise using its **power spectral density (spectrum)**.
- This shows how much power (energy) the noise has at each frequency.

Example : Voice contains frequencies from 20 Hz to 20 kHz.

- A microphone captures the sound.
- A band-pass filter lets only 1 Hz around a chosen frequency pass.
- A power meter measures the signal's power.
- Sweeping the filter from 20 Hz to 20 kHz gives the **voice spectrum**, $S(f)$.



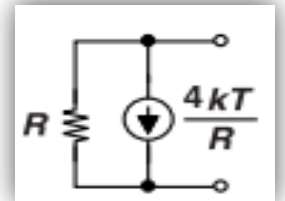
Device Noise



- In CMOS, noise mainly comes from resistors (thermal) and MOSFETs : thermal & flicker ($1/f$).

Resistors

- Convert thermal energy into voltage across it.
- Spectrum of the voltage is given by : $S(f) = 4kTR \text{ V}^2/\text{Hz}$
- The noise is modeled as a voltage source in series with the resistor.
- Also called the “**white noise**” as it carries equal power at all frequencies.
- Sometimes modeled as a parallel current source: $S(f) = 4 k (T/R) \text{ A}^2/\text{Hz}$



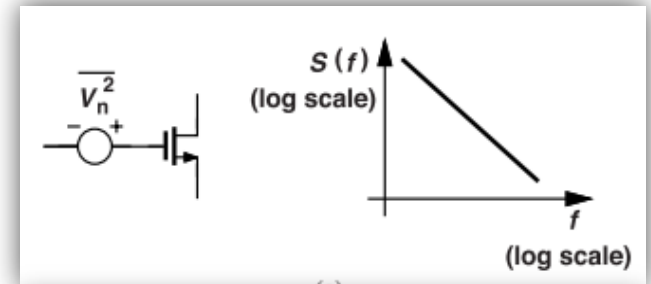
Why you can't write $V_n = \sqrt{4kTR}$.

- $4kTR$ tells us how much **noise power** (in volts^2) is present in **each 1 Hz** slice of frequency. It doesn't give actual power.

MOSFET :

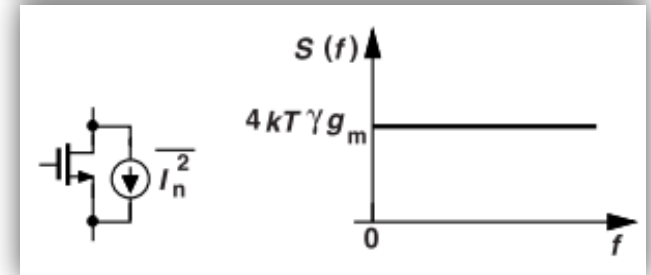
1. Flicker noise (1/f)

- A type of voltage noise that affects MOSFETs at low frequencies
- Modeled as Voltage source in series with the gate.
- Spectrum of the voltage :
 - $S(1/f)(f) = K / (W * L * C_{ox} * f) V^2/Hz$
 - K is a process-dependent parameter



2. Thermal noise

- The thermal noise of MOSFETs in the saturation region.
- Modeled as a Current source parallel with drain and source terminals.
- Spectrum of the voltage :
 - $S(f) = 4kT\gamma g_m A^2/Hz$
 - γ : Excess noise coefficient ($\approx 2/3$ to 1)
 - Higher γ = more thermal noise.. Compared to ideal model



Propagation of Noise

Q. How does Noise generated at the source affect the output ?

- ***Steps to Find out the propagation of noise :***

1. Find Transfer Function :

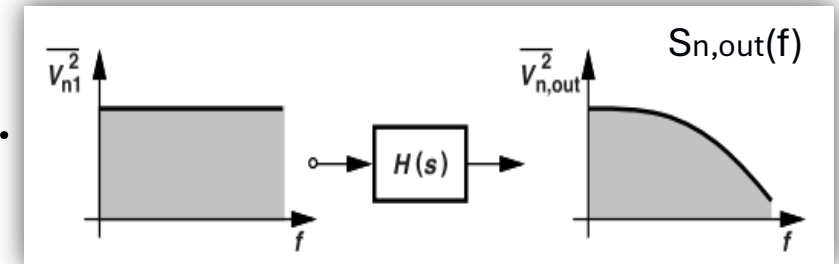
- get $H(s)$ from V_{n1} to $V_{n,out}$.

2. Frequency Domain :

- put $s=j\omega$; get $H(f)$; Magnitude $|H(f)|^2$.

3. Get Output Noise Spectrum :

- $S_{n,out}(f) = S_{n1}(f) * |H(f)|^2$



- $S_{n1}(f)$ denotes the spectrum of V_{n1}
- $|H(f)|^2$ shaped the noise signal

Example : If $S_{n1}(f)$ is white and $H(s)$ a low-pass filter.

Power & Spectral Density

Average Power of Noise

1. For a random signal, $x_2(t)$, we define the average power as :

- Integrate its squared value over the time.
- T denotes the period

$$P_{x_2} = \overline{x_2^2(t)} = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{+T/2} x_2^2(t) dt.$$

2. The average power can also be obtained from the spectrum of the signal.

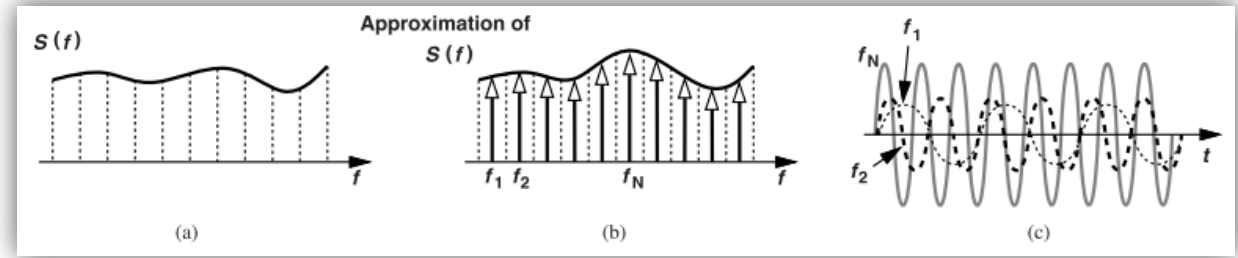
- Integrate its value over the freq range.

$$P_{x_2} = \int_{-\infty}^{+\infty} S_{x_2}(f) df.$$

Parseval's theorem : Total average power = Area under the Spectral Density Curve.

If a frequency has power in the spectrum, it shows up as part of the random waveform in time.

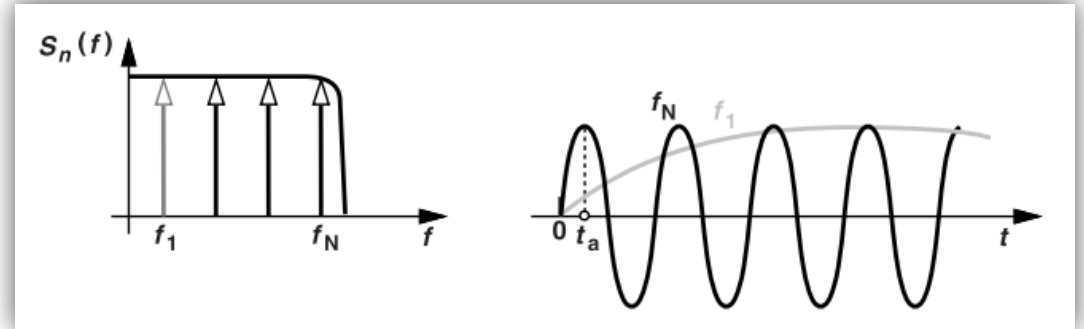
Approximation of Noise Spectrum



- Noise with any spectrum can be approximated by breaking it into many narrow frequency bands(called slices /slices) .
- Each band is represented as an impulse, which corresponds to a sinusoid in the time domain.
- So, the noise is modeled as the sum of many sinusoids at different frequencies.
- But this is only for short period of times that match the bandwidth of each frequency slice.
- **EXAMPLE:** If each slice is 1 kHz wide (in freq domain), the sinusoid model is only accurate for about 1 millisecond
- Beyond that, the sum of sinusoids can't fully capture the randomness.
- To model noise accurately over longer times, we use narrower slices.

Accumulation of Noise with Time

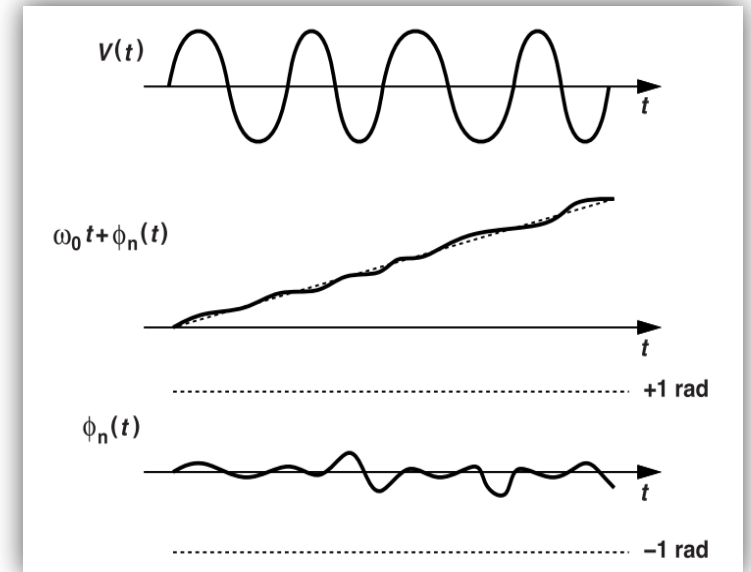
- When looking at noise over a short "observation window" of **[0 to t_a]**,
 - Only the fast-changing (high-frequency : f_N) noise components matter.
 - Slow noise (low-frequency: f_1) doesn't change much during that window, so it affects the signal less.



Observing noise for t_a seconds is like applying a **high-pass filter** that blocks very slow noise variations and lets through only high-freq noise.

Basic Jitter and Phase Noise Intro

- A perfect oscillator has fixed timing, but real ones have **noise** that causes **random shifts in phase** called **jitter** and **phase noise**.
- We model this as: $V_{out}(t) = V_0 * \cos(\omega_0 * t + \phi_n(t))$
 - where $\phi_n(t)$ is the excess phase (noise induced)
 - ω_0 is called as the "carrier" frequency. (ideal)
- **EXAMPLE:**
- Plot $V_{out}(t) = V_0 * \cos(\omega_0 * t + \phi_n(t))$ if $\phi_n(t) \ll 1$ rad.
 - Total phase is the entire argument inside the $\cos()$
 - Excess phase is the phase in addition to the $\omega * t$ term.
 - Since $\phi_n(t)$ is way lesser than 1, it causes mild disturbances in the straight-line waveform of $\omega * t$



Jitter : Definition

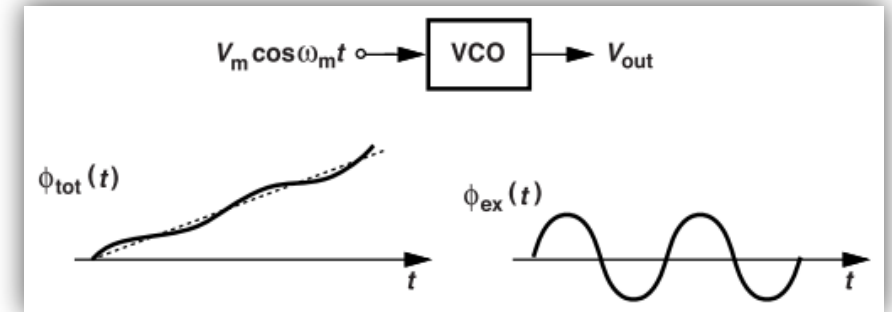
- As discussed before Jitter is deviation of zero-crossings of a periodic signal from its ideal points. And it is a Time-domain phenomenon.

- Example : Can VCO's exhibit Jitter ??**

- Input : $V_{\text{cont}}(t) = V_m \cos(\omega_m t)$

- Output $V_{\text{out}}(t)$
$$= V_0 \cos \left(\omega_0 t + \frac{K_{\text{VCO}} V_m}{\omega_m} \sin \omega_m t \right)$$

- Observation :** *The maximum departure of the zero crossings from their ideal points in time occurs when the excess phase reaches a maximum or minimum.*

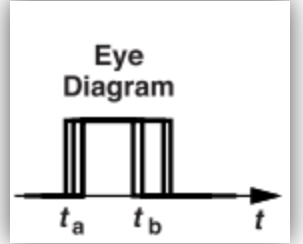


- Deterministic jitter :** Arises if the osc freq (ω) is periodically disturbed.
- Peak to Peak Jitter = $(2 * K_{\text{vco}} * V_m) / \omega_m$ radians = $(2 * K_{\text{vco}} * V_m) / (\omega_m * \omega_0)$ sec

{Note : if we increase ω_m ; we reduce the Jitter}

Visualizing and Measuring Jitter: The Eye Diagram

- Take multiple cycles of the “jittery” clock signal and stack them on top of each other.
- This is done by using an ideal clock as a reference and overlaying each measured cycle.
- The wider the eye opening, the less jitter; a smaller opening means more jitter.
- *An eye diagram is a powerful way to see timing jitter, but frequency analysis is needed to know if the noise is random/deterministic.*



Jitter types : Random and Periodic fluctuations in the signal

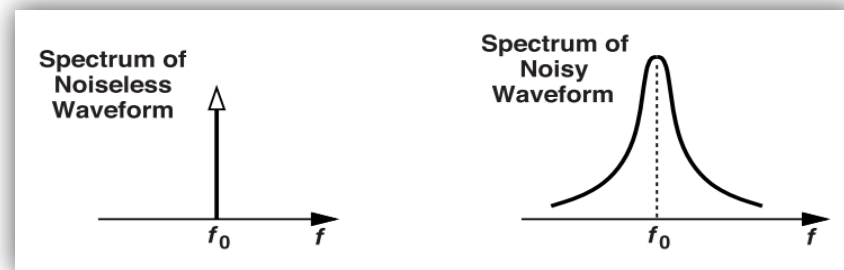
Phase Noise: Definition & Spectra

Phase noise : Random fluctuations in the phase of an osc output.

- It is analysed in freq domain only.
- $V_{out}(t) = V_0 * \cos(\omega_0 * t + \phi_n(t))$ where $\phi_n(t)$ is called as Phase noise.

Effect on Spectrum

- Ideal case : The spectrum is just a single spike (impulse) at the carrier frequency (f_0).
- Phase noise spreads the energy, so the spectrum is no longer just at f_0 , but is “broadened” around it. *{Why? frequency randomly jumps around f_0 due to noise}*



Relationship b/w Phase Noise Spectrum and Output Spectrum

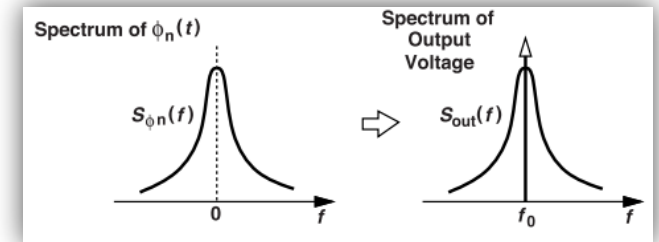
$$V_{out}(t) = V_0 \cos[\omega_0 t + \phi_n(t)] \approx V_0 \cos \omega_0 t - V_0 \phi_n(t) \sin \omega_0 t.$$

Assume $\phi_n(t) \ll 1$ rad

1. Carrier ω_0 is responsible for the shift of $\phi_n(t)$ spectrum from 0 to f_0 in the output voltage spectrum.
2. Farther you go from f_0 , less likely you are to see large deviations.
 - Spectrum Tappers off.

How do we measure Phase noise ?

1. Pick a frequency offset ; (100kHz away from f_0).
2. Measure the noise power in a very narrow range (1 Hz wide) at that offset.
3. Compare this noise power to the main signal's power (the carrier power).
4. Express the result in dBc/Hz (decibels relative to the carrier, per Hertz of bandwidth).



Jitter vs. Phase Noise

- For **deterministic Jitter** , we use peak-to-peak jitter i.e., $(2 \cdot K_{vco} \cdot V_m) / \omega_m$ radians.
 - This is 4 times the normalized Sideband Magnitude (as discussed before).
- For **random jitter**, we use RMS value
 - Peak-to-peak jitter only makes sense for repetitive, bounded patterns, but random jitter can, in theory, be infinitely large.
 - RMS jitter gives a statistical measure of the average size of timing variations, making it practical for real systems.

Calculate RMS Jitter:

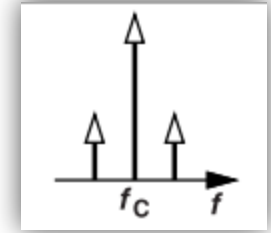
- RMS Jitter = the square root of the area under the phase noise spectrum.

$$J_{rms} = \sqrt{\int_{-\infty}^{+\infty} S_{\phi n}(f) df}.$$

$$\overline{\phi_n^2(t)} = \int_{-\infty}^{+\infty} S_{\phi n}(f) df.$$

Parseval's theorem that the average power is equal to the area under the spectrum:

Types of Jitter : In detail

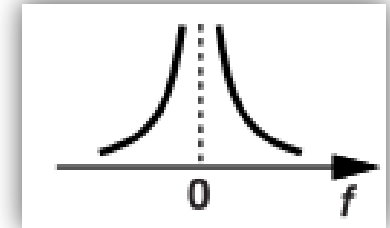


1. Deterministic Jitter :

- Caused by predictable, periodic effects. (Due to Periodic Modulation)
- Seen as sidebands in the frequency spectrum.
- Bounded (has peak-to-peak value)

2. Absolute Jitter :

- Caused by **low-frequency noise**
- No external reference, so error **accumulates over time**.
- Grows **without limit — unbounded**.



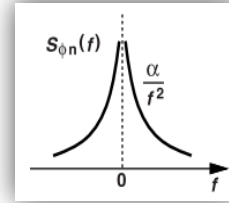
3. Cycle-to-Cycle Jitter :

- Measures the period changes from one cycle to the next : **Jcc=T_k-T_{k-1}**.
- Bound and Measurable
- For many Cycles , rms value of Jcc :

$$J_{cc,rms} = \lim_{n \rightarrow \infty} \sqrt{\frac{1}{n} \sum_{k=1}^n (T_k - T_{k-1})^2}.$$

- Only high-frequency noise affects this, so it remains small and bounded.

Example : White noise Induced Phase noise $S_{\phi n}(f)$:



- Assume $S_{\phi n}(f) = \alpha/f^2$; Power declines as $|f|$ increases.
- If this phase noise affect a 1GHz o/p of Oscillator, only the higher frequency components will affect the output.
- So, the white noise induced phase noise is (ω_0 is osc freq) :
- We simplify the equation to

$$J_{cc,rms}^2 = \frac{4\pi}{\omega_0^3} \omega^2 S_{\phi n}(\omega),$$

$$J_{cc,rms} = \sqrt{\frac{2\alpha}{f_0^3}}.$$

and finally to this :

$$\frac{J_{cc,rms}}{T_0} = \sqrt{\frac{2\alpha}{f_0}}.$$

4. Period Jitter (also called Cycle jitter):

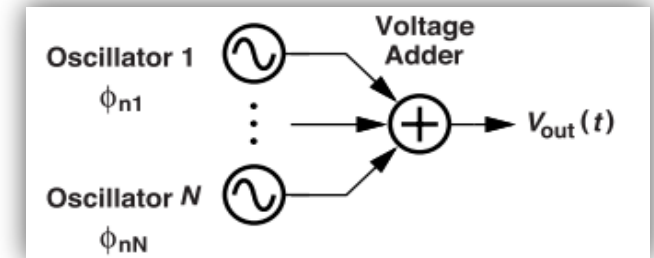
- Random departures of the period from its the average value, T_{avg} .
- **$J_c = J_{cc} / \sqrt{2}$** for white-noise-induced jitter. Here,
- Bounded

$$J_c = \lim_{n \rightarrow \infty} \sqrt{\frac{1}{n} \sum_{k=1}^n (T_k - T_{avg})^2}.$$

Trade-Offs b/w Phase noise and Power

- Power and phase noise are Inversely related.

EXAMPLE: Add N-identical oscillators to an Adder.



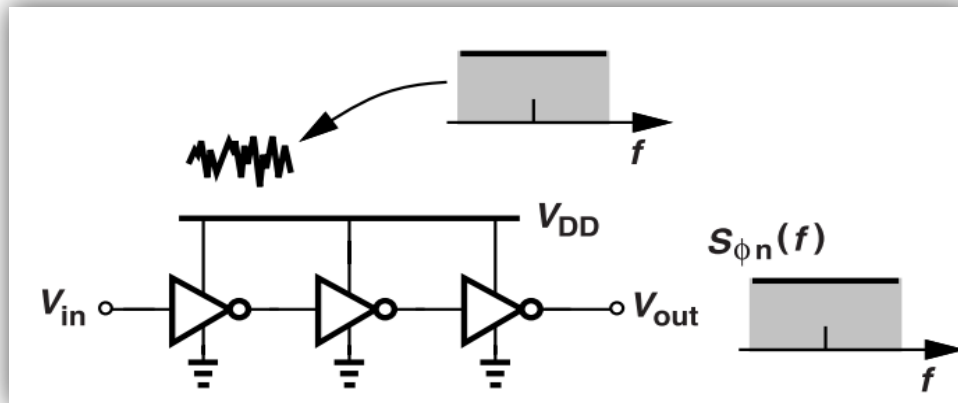
- Output of 1-osc: **$V_{out}(t) = V_0 \cos \omega_1 t - V_0 \phi_{n1} \sin \omega_1 t$**
- Output of N-osc : **$V_{out}(t) = N(V_0 \cos \omega_1 t) - V_0(\phi_{n1} + \dots + \phi_{nN}) \sin \omega_1 t$** .
- The normalized phase noise is equal to **$(\phi_{n1} + \dots + \phi_{nN})/N$** and exhibits a spectrum given by : **$S_{\phi}(f) = S_{\phi_{n1}} + \dots + S_{\phi_N} / N^2$**
- N-osc phase noise spectra are equal and uncorrelated; **$S_{\phi}(f) = S_{\phi_{n1}} / N$**
 - ✓ Phase noise is reduced by N times But power increases by N^2 times
- This method doesn't apply for deterministic jitter.

Basic Phase Noise Mechanisms

1. Phase noise vs frequency noise

a) Delay Line (chain of inverters) :

- Delay of the chain of inverters depends on the supply voltage V_{DD} .
- ΔV change in V_{DD} creates Δt_D in the line's delay and $\Delta \phi$ phase shift.
- Supply noise is V_n , the output of the delay line is expressed as :
 - $V_{out} = \{V_{DD}/2\} + \{V_{DD}/2\} * \cos(\omega_{int} + \phi_0 + K_{DD} * V_n)$
 - Phase shift due to Supply noise is : $(K_{DD} * V_n)$



- Since, phase shift $K_{DD} * V_n$ is linear, white supply noise directly changes the output phase resulting in the white phase noise

b) Ring Oscillators:

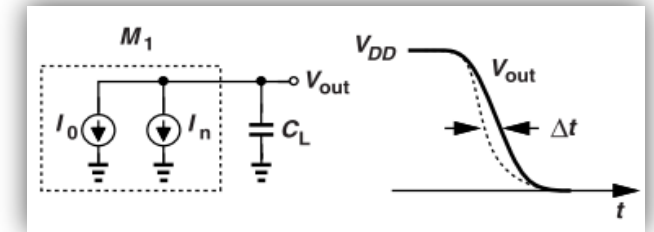
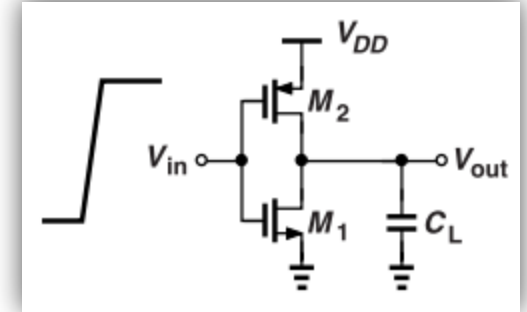
- VDD in ring Oscillators affect both frequency and phase.
- Higher VDD = Higher Oscillation frequency.
- **white supply noise translates to white frequency noise**
- Gain from Vdd is $K_{DD} = \partial\omega_{out}/\partial V_{DD}$;
- The oscillator output is :
$$V_{out} = \frac{V_{DD}}{2} + \frac{V_{DD}}{2} \cos(\omega_{osc}t + \int K_{DD} V_n dt).$$

If length of delay line made infinite ; will its phase noise match that of a ring oscillator. ???

- In a delay line, the phase noise is added linearly since supply noise directly affects phase noise.
- In ring oscillators, noise changes the oscillation frequency, which then accumulates over time into phase noise.
- The assumption misses the integration effect in oscillators.

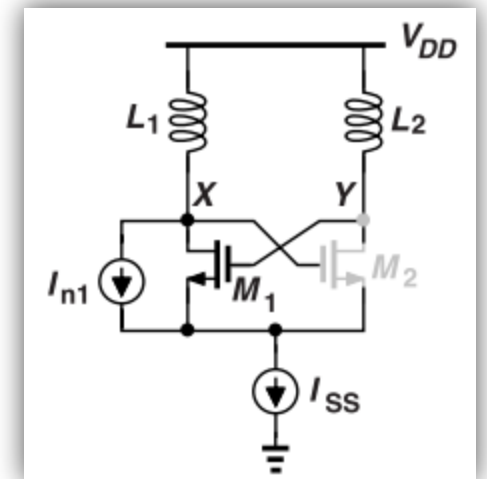
2. Ring oscillators : Jitter & Phase noise

- Input rises from 0 to V_{DD} .
 - M_1 : turns on and discharges through C_L . { V_{DD} to 0}
 - I_n : Noise from M_1 : flows through C_L Causing Jitter Δt .
 - White current noise = white phase noise for Inverter.
- For N-cascaded Inverters:
 - Phase noise rises by N factor
 - RMS jitter by root(N) factor



3. LC Oscillators : Jitter & Phase noise

- **I_{n1}** (thermal noise from M_1)
- injects jitter into the LC tank, but only when **M_1 is ON**.
- When **M_2 is OFF**, **I_{SS} blocks I_{n1}**
- So, noise reaches the tank **only during part of the cycle**.
Thus, its effect on phase is **time-dependent**.



Effect of Jitter & PN on Performance

Jitter

- Affects data sampling, especially in high-speed links (e.g., processor to memory).
- Ideal clock samples correctly; jittered clock may cause logic errors or metastability.
- Safe limit: Clock jitter must be less than half the data period.

Phase noise

- Interferes with Nearby Channels : Noise spreads out → affects weaker signals on adjacent frequencies.
- Corrupts Modulated Signal : Adds random phase error → causes symbol decoding issues (e.g., in WiFi).
- Reciprocal Mixing in RX: Strong nearby signals leak into desired signal due to phase noise.